Git and Github introduction

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What is git and github?

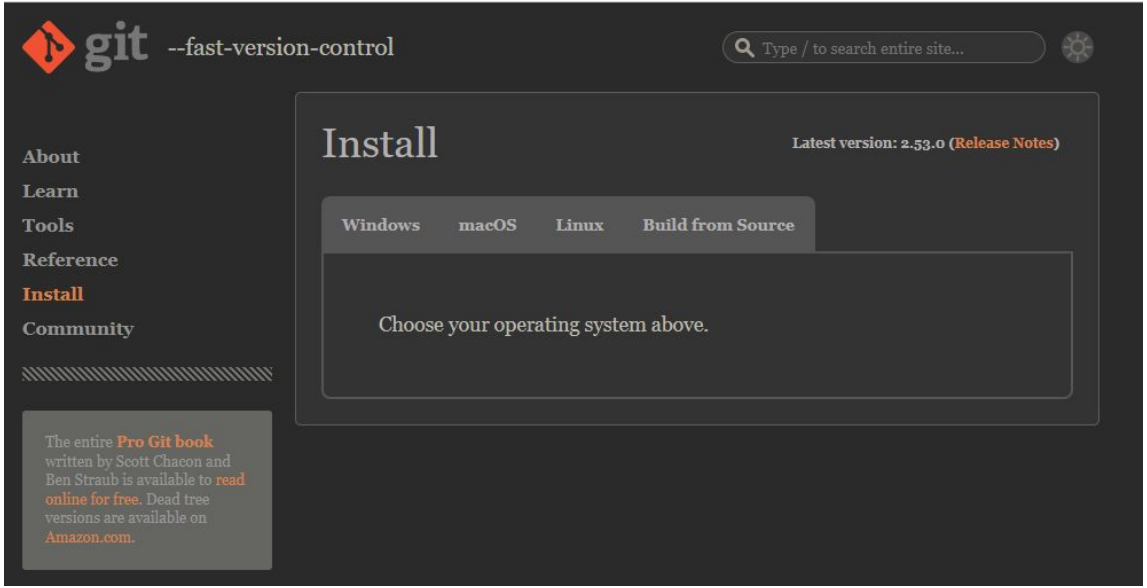
Git is a version controller, it controls all the changes in your code!
If you make a mistake you can go back to a previous version of your code before the mistake!

Github stores your code and all the changes that were made to it!
Github is a website that stores all the code so you can collaborate on it with multiple people at the same time using git!

How do I install and setup git?

Install it it's free!

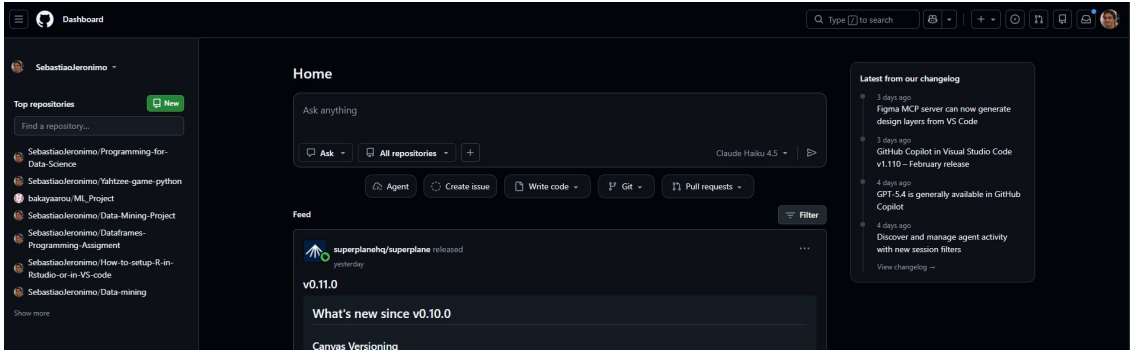
I will leave the link in the comments -> https://git-scm.com/install/



How do I setup github?

To use github you can just create an account its free!

The link will also be in the comments -> https://github.com/



Lets create a repository

After having git installed (confirm with git --version)

And created a github account

Lets create a github repository!

But what is a repository?
A repository it's just a place to store our code so we can work with it using git , this is much better than working locally because we can store our code revert mistakes and work with others at the same time!

Using git commands for the first time

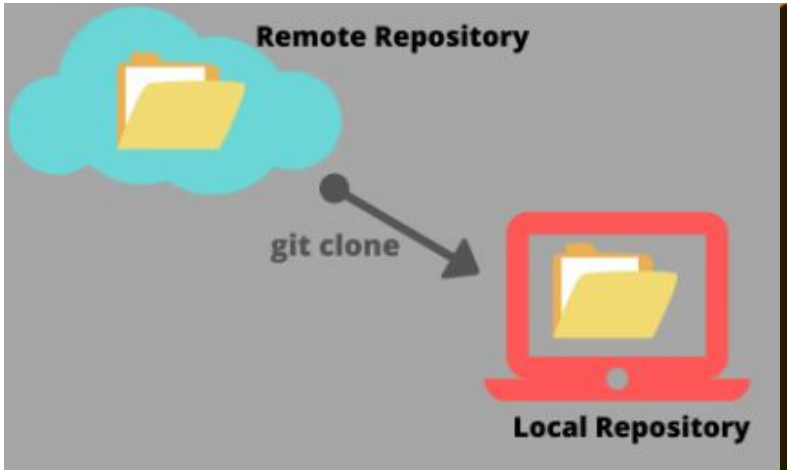
`git clone`	Bringing the project from the github repository to your computer.
`git add <specific file>`	Choosing the changes that you want to do to the code.
`git commit -m “note”`	Confirming the changes with a note.
`git push`	Sending the code with your updated changes to the github repository.

Resolving conflicts

`git pull`	When you don't have the last version you can pull the changes that you are missing.

Git clone

Cloning a repository is getting the files that are stored on the github repository (location in the cloud) to your computer.




```
# Syntax
$ git clone <repository>

# Example
$ git clone https://github.com/username/myproject.git
```

Git Add

We can use the git add to choose the changes that we want to add to the repository.

How to Use git add

Add a Single File

```
git add filename.txt
```

Add All Files in the Directory

```
git add .
```

Add Multiple Files

```
git add file1.txt file2.txt
```

Add All Files

```
git add -A
```

Git Commit

We use git commit to confirm the changes we want to add to the repository

When we commit to the repository we must add a note. (-m <message>)

Syntax:

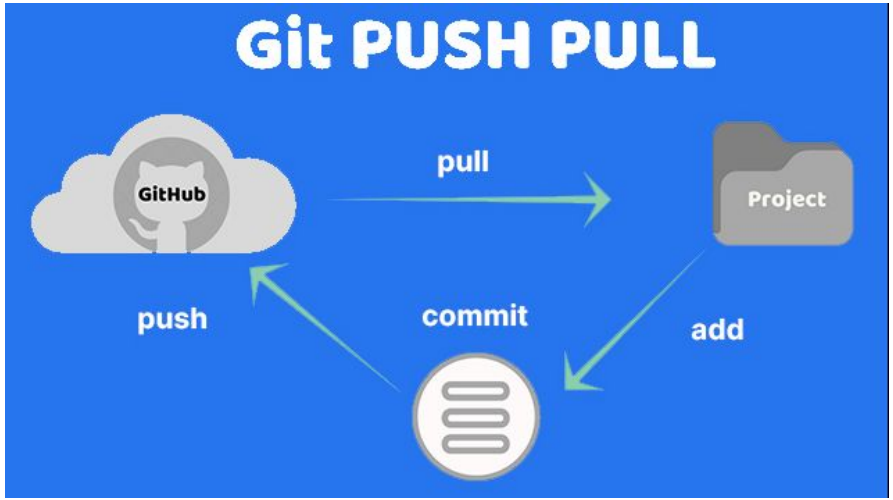
```
git commit -m "message"
```

Message example:

```
git commit -m "First release of Hello World!"
```

Git Push

After confirming the changes with git commit we will send the changes to the repository with git push.



Syntax:

```
git push
```

Resolving conflicts with Git Pull

When multiple developers are working:

Commit changes locally

```
git commit -m "message"
```

Push changes to remote

```
git push origin branch-name
```

Pull latest changes before pushing again

```
git pull
```

Resolve conflicts if needed

```
Merge conflicts may occur if multiple people edited the same files.
```

Re-push after resolving conflicts

```
git push
```

Before pushing you have to pull the latest changes if you don't have the updated version

Syntax:
Git pull